Supply Chain Analytics - Homework Week 1

Forecasting Airline Passengers

You work for a competitive airliner, which has 50 planes in operation, of which 8 are in repair now, and for some others maintenance in planned, but all planes should be operational before summer—last year, all planes were needed to fulfill the demand in one month. Moreover, last year, a record-breaking number of 4.6 million passengers were served, which is the result of a bumpy but steady growth over the last decade.

Now, since you are responsible for customer insights, you are asked to help the board provide insights on the number of airline passengers in the upcoming year. The board is highly curious to know these values for the next year (per month) so as to adjust their fleet and staffing plans. Since customer satisfaction is most important, they want to be able to meet the peak demand with 95% probability. They also want your opinion on the longer horizon (i.e., the next 5 years) to know whether further strategic investments are necessary.

The report should be presentable and cover the following aspects (the code should be provided too). Note you may use Python packages and I advise you to consider `statsmodels`: `https://www.statsmodels.org/stable/tsa.html`.

I. Visualize the data, check for strange values, and identify trends or seasonality visually.

II. Stationarity testing: Test the stationarity of the time series using appropriate tests such as the Augmented Dickey-Fuller (ADF) test, the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test, or the Phillips-Perron test.

III. Time Series Modeling: Build (a) an exponential smoothing model and (b) an (S)AR(I)MA model. To select the right (hyper)parameters, one can use hit-and-trial, a Box-Jenkins approach, a grid search, or optimization.

IV. Model evaluation: Evaluate the performance of the two models using appropriate metrics (bias, root mean squared error (RMSE), mean absolute error (MAE), and mean absolute percentage error (MAPE)).

V. Forecasting: Use the trained models to forecast the sales for the remainder of the year and visualize the forecasted values along with the historical values. Provide a *relevant* prediction interval (if difficult, use simulation wisely).

VI. Advising: Do the models align with one another? Advise management regarding the *next* year and the *longer* horizon, say 5 years. Support your arguments with analytics.

Deadline: see Canvas.

Presentation: Make sure the work is presentable, i.e., understandable step-by-step. As an example, a PowerPoint or a Jupyter notebook will be acceptable submission types.

Good luck.